

Clinical Manifestations

HISTORY

Symptoms begin several days to several weeks after ingestion of contaminated water or food. Common early symptoms include headache, fever, generalized aching, abdominal discomfort, cough, and constipation. Delirium or mental torpor may occur, particularly with high fever. The heart rate may be relatively bradycardic for the degree of fever (pulse-temperature dissociation). Symptoms progressively worsen, and fever typically reaches $39.4^{\circ}\mathrm{C}$ to $40.6^{\circ}\mathrm{C}$ ($103^{\circ}\mathrm{F}$ to $105^{\circ}\mathrm{F}$) by the end of the first week. During the second week, rose spots may appear on the trunk, and diffuse abdominal cramping becomes prominent, sometimes accompanied by diarrhea.

CUTANEOUS LESIONS

After 7 to 10 days of high fever, **rose spots**—the classic skin lesions of systemic *Salmonella* infection—may develop. These are 2- to 3-mm, slightly raised, nontender, pink maculopapular lesions that blanch with pressure. They typically appear in crops of 10 to 20 lesions on the anterior trunk, most commonly between the nipple area and the umbilicus, and rarely involve the back or extremities.

Without antimicrobial treatment, the spots usually turn brownish as they fade and disappear within 3 to 4 days. New crops may continue to emerge over the next 2 to 3 weeks in untreated cases. Rose spots are observed in approximately 63% of untreated typhoid fever cases but occur less frequently in enteric fevers caused by non-typhoidal *Salmonella* species. Early antibiotic therapy is associated with a decreased incidence of rose spots. Detection may be more difficult in individuals with darker skin, which may contribute to underreporting in people of color.

Other cutaneous findings during acute enteric fever include: - **Erythema typhosum**: an erythematous, confluent rash occurring in the first week. - **Erythema nodosum** and **urticarial lesions**. - Rare reports of subcutaneous abscesses and skin ulcers, particularly from the Indian subcontinent. - Herpes labialis is distinctly uncommon in enteric fever.

OTHER PHYSICAL FINDINGS

While cutaneous lesions are present, patients may exhibit disorientation and signs suggestive of pneumonia. The abdomen is often distended, tympanic, and diffusely tender, with possible localization to the right lower quadrant. Splenomegaly is common but may be difficult to palpate due to abdominal distension and guarding.

Laboratory Findings

- **Leukopenia** is typical, though white blood cell counts can range from 3,000 to 20,000/mm³.
- The proportion of mononuclear cells may be increased, and atypical lymphocytes may be seen—findings that can mimic viral illness.
- **Thrombocytopenia** may occur; hemolysis is rare.
- During the first week of illness, **blood cultures** are usually positive for *Salmonella*.
- By the second week or at the onset of diarrhea, **stool cultures** become positive.
- **Isolation of *Salmonella enterica* serotype Typhi from blood** remains the gold standard for diagnosis of typhoid fever.

Histopathology

Histologic examination of a rose spot shows marked dilatation of dermal capillaries, significant edema, and dense pericapillary infiltration by macrophages. *Salmonella* organisms may be identified within these macrophages.

Diagnosis and Differential Diagnosis

Diagnosis of *Salmonella* enteric fever can be challenging, especially in sporadic cases where the characteristic rash and gastrointestinal symptoms have not yet developed and there is no clear history of travel to an endemic area. Early manifestations such as headache, cough, and high fever are nonspecific and may resemble other febrile illnesses, including viral infections, malaria, or other bacterial sepsis syndromes. Clinical suspicion should remain high in the context of prolonged fever and relative bradycardia, particularly in patients with potential exposure.